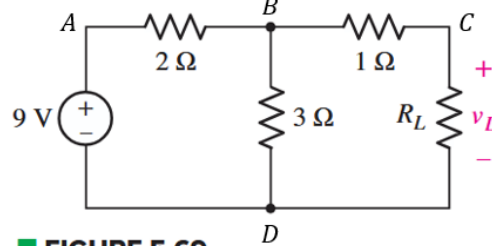


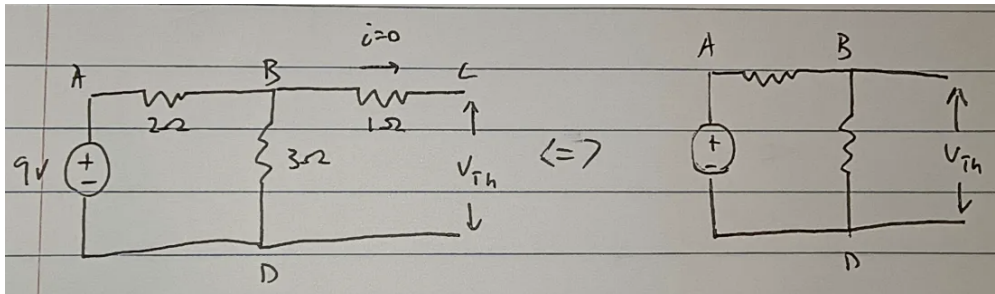
HW 4 - Zeshui Song

- 5.25** 25. (a) Referring to Fig. 5.69, determine the Thévenin equivalent of the network connected to R_L . (b) Determine v_L for $R_L = 1\ \Omega$, $3.5\ \Omega$, $6.257\ \Omega$, and $9.8\ \Omega$.



■ FIGURE 5.69

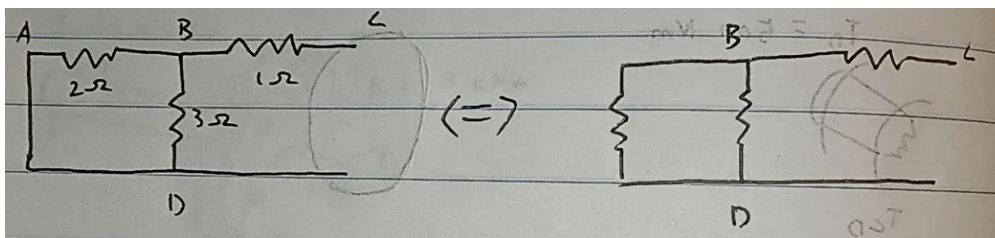
(a) Finding the Thévenin equivalent. Starting with finding the open circuit voltage V_{oc} across the load resistor R_L (remove R_L from the circuit):



Thus, V_{oc} is the same voltage across nodes B and D. Using voltage division:

$$V_{BD} = V_{oc} = \frac{[3\ \Omega]}{[2 + 3\ \Omega]}[9\ \text{V}] = 5.4\ \text{V}$$

To find the Thévenin equivalent resistance R_{Th} , we turn off all independent sources (replace voltage sources with **short circuits** and current sources with **open circuits**) and find the equivalent resistance across nodes C and D:



Add the parallel resistors $2\ \Omega + 3\ \Omega$:

$$R = \frac{[2 \cdot 3\ \Omega]}{[2 + 3\ \Omega]} = 1.2\ \Omega$$

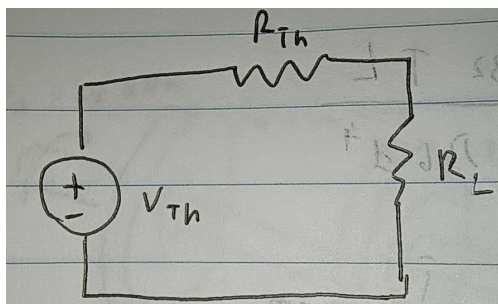
Add the series resistor $1\ \Omega$:

$$R_{Th} = 1\ \Omega + 1.2\ \Omega = 2.2\ \Omega$$

Thus:

$$\boxed{V_{Th} = V_{oc} = 5.4\ \text{V} \quad R_{Th} = 2.2\ \Omega}$$

(b) Using the Thevenin equivalent circuit, we can use voltage division to find the voltage across the load resistor R_L :



- $R_L = 1\Omega$:

$$V_L = \frac{[1\Omega]}{[2.2\Omega + 1\Omega]}[5.4 \text{ V}] = 1.687 \text{ V}$$

- $R_L = 3.5\Omega$:

$$V_L = \frac{[3.5\Omega]}{[2.2\Omega + 3.5\Omega]}[5.4 \text{ V}] = 3.315 \text{ V}$$

- $R_L = 6.257\Omega$:

$$V_L = \frac{[6.257\Omega]}{[2.2\Omega + 6.257\Omega]}[5.4 \text{ V}] = 3.995 \text{ V}$$

- $R_L = 9.8\Omega$:

$$V_L = \frac{[9.8\Omega]}{[2.2\Omega + 9.8\Omega]}[5.4 \text{ V}] = 4.41 \text{ V}$$

5.30 30. (a) Employ Thévenin's theorem to obtain a simple two-component equivalent of the circuit shown in Fig. 5.72. (b) Use your equivalent circuit to determine the power delivered to a $100\ \Omega$ resistor connected to the open terminals. (c) Verify your solution by analyzing the original circuit with the same $100\ \Omega$ resistor connected across the open terminals.

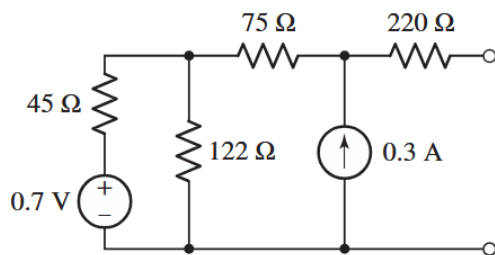
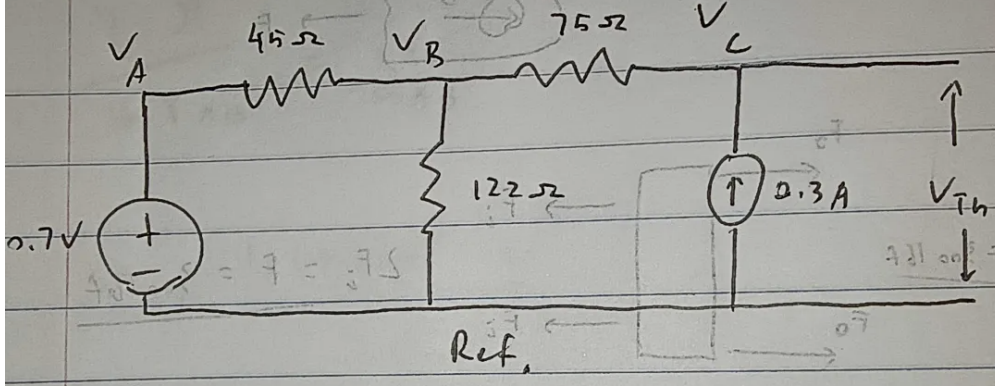


FIGURE 5.72

(a) Finding the Thevenin equivalent. Starting with finding the open circuit voltage V_{oc} by using nodal analysis. Like before, there is no current flowing through the 220Ω resistor, so the voltage across the current source is equal to V_{oc} :



Node A:

$$V_A = 0.7 \text{ V}$$

KCL at node B:

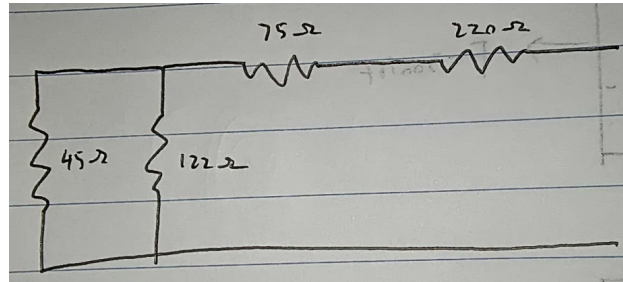
$$\frac{[V_A - V_B]}{[45 \Omega]} = \frac{V_B}{122 \Omega} + \frac{V_B - V_C}{75 \Omega} \implies 0.0437522V_B - 0.0133333V_C = 0.015555$$

KCL at node C:

$$\frac{V_B - V_C}{75 \Omega} = -0.3 \text{ A} \implies 0.0133333V_B - 0.0133333V_C = -0.3$$

Thus, $V_B = 10.373 \text{ V}$ and $V_C = 32.873 \text{ V}$. Therefore, $V_{oc} = V_C = 32.873 \text{ V}$.

To find the Thevenin equivalent resistance R_{Th} , we turn off all independent sources:



Combine the parallel resistors 45Ω and 122Ω :

$$R = \frac{[45 \cdot 122 \Omega]}{[45 + 122 \Omega]} = 33.87 \Omega$$

Add to the series resistors 75Ω and 220Ω :

$$R_{Th} = 33.87 \Omega + 75 \Omega + 220 \Omega = 327.87 \Omega$$

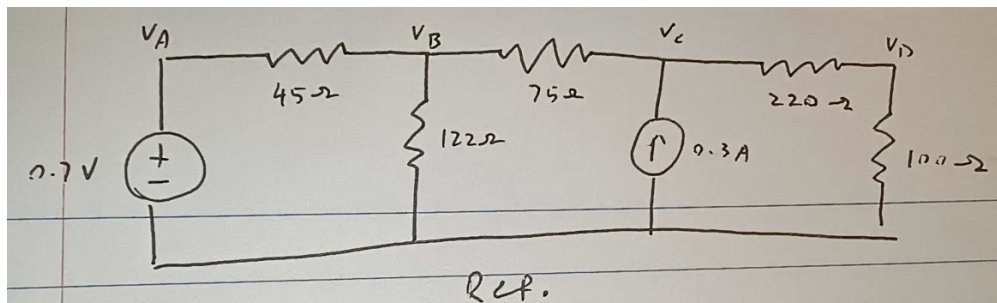
(b) Power delivered to the load resistor $R_L = 100\Omega$. First find the voltage across R_L using voltage division:

$$V_L = \frac{[100 \Omega]}{100 + 327.87 \Omega} [32.873 \text{ V}] = 7.682 \text{ V}$$

Thus the power delivered to R_L is:

$$P_L = \frac{V_L^2}{R_L} = \frac{[7.682 \text{ V}]^2}{[100 \Omega]} = \boxed{0.59 \text{ W}}$$

(c) We can apply nodal analysis to the circuit including the 100Ω load resistor to find the voltage across R_L .



Node A:

$$V_A = 0.7 \text{ V}$$

KCL at node B:

$$\frac{[V_A - V_B]}{[45 \Omega]} = \frac{V_B}{122 \Omega} + \frac{V_B - V_C}{75 \Omega} \implies 0.0437522V_B - 0.0133333V_C = 0.015555$$

KCL at node C:

$$\frac{V_B - V_C}{75 \Omega} = [-0.3 \text{ A}] + \frac{V_C - V_D}{220 \Omega} \implies 0.0133333V_B - 0.017878V_C + 0.004545V_D = -0.3$$

KCL at node D:

$$\frac{V_C - V_D}{220 \Omega} = \frac{V_D}{100 \Omega} \implies 0.004545V_C - 0.014545V_D = 0$$

Thus, $V_B = 7.848 \text{ V}$, $V_C = 24.586 \text{ V}$, and $V_D = V_L = 7.682 \text{ V}$. Thus the power delivered to R_L is:

$$P_L = \frac{V_L^2}{R_L} = \frac{[7.682 \text{ V}]^2}{[100 \Omega]} = \boxed{0.59 \text{ W}}$$

- 5.32** 32. Determine the Thévenin equivalent of the network shown in Fig. 5.74 as seen looking into the two open terminals.

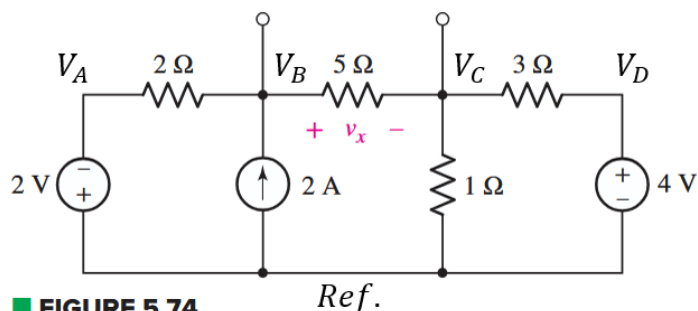


FIGURE 5.74

Ref.

The open circuit voltage $V_{oc} = V_B - V_C$. Using nodal analysis:

Node A:

$$V_A = -2 \text{ V}$$

KCL at node B:

$$\frac{V_A - V_B}{[2 \Omega]} = \frac{V_B - V_C}{[5 \Omega]} - [2 \text{ A}] \quad \Rightarrow \quad 0.5V_A - 0.7V_B + 0.2V_C = -2$$

KCL at node C:

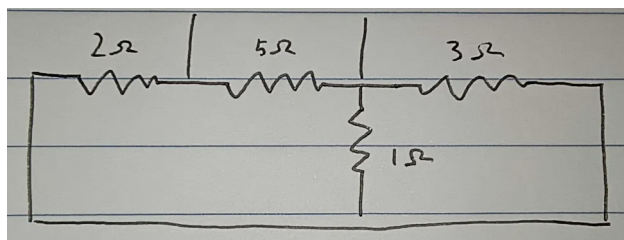
$$\frac{V_B - V_C}{[5 \Omega]} = \frac{V_C}{[1 \Omega]} + \frac{V_C - V_D}{[3 \Omega]} \quad \Rightarrow \quad 0.2V_B - 1.5333V_C + 0.333V_D = 0$$

Node D:

$$V_D = 4 \text{ V}$$

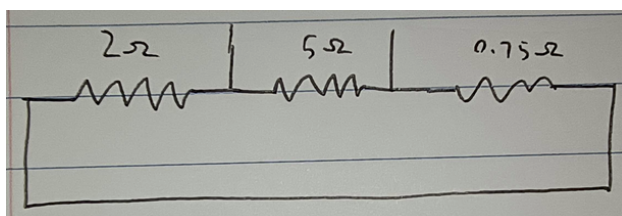
Thus, $V_B = 1.741 \text{ V}$ and $V_C = 1.096 \text{ V}$. Therefore, $V_{oc} = V_B - V_C = 0.645 \text{ V}$.

Finding the Thevenin equivalent resistance R_{Th} , we turn off all independent sources:



Combine the parallel resistors $1 \Omega + 3 \Omega$:

$$R = \frac{[1 \cdot 3 \Omega]}{[1 + 3 \Omega]} = 0.75 \Omega$$

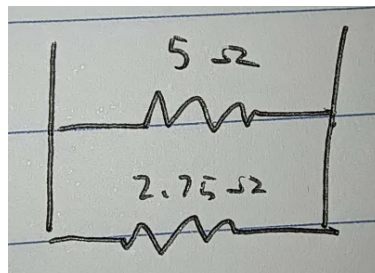


Combine 2Ω and 0.75Ω in series:

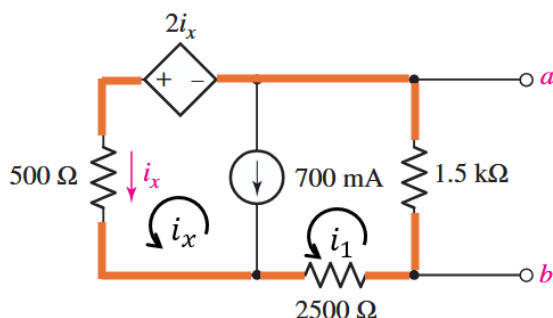
$$R = 2\Omega + 0.75\Omega = 2.75\Omega$$

Finally, combine 2.75Ω and 5Ω in parallel:

$$R_{Th} = \frac{[2.75 \cdot 5\Omega]}{[2.75 + 5\Omega]} = 1.774\Omega$$



5.40 40. Determine the Norton equivalent of the circuit drawn in Fig. 5.81 as seen by terminals a and b . (There should be no dependent sources in your answer.)



■ **FIGURE 5.81**

The *Norton equivalent* circuit is the same as the Thevenin equivalent circuit, except the voltage source is transformed into a current source. However, since there is a dependent voltage source in the circuit, we cannot directly find the Thevenin equivalent resistance by turning off all independent sources and summing up the resistances. Instead, we define a short circuit current $i_{sc} = i_N$ by shorting the terminals of the load (place a wire across it). Then, the two circuits are related by a source transformation:

$$V_{Th} = R_{Th}i_N$$

$$R_{Th} = R_N$$

Note: If the dependent source is controlled by the load voltage, shorting it would cause this voltage to be zero and thus the dependent source would be turned off.

First, using mesh analysis, we can find the current i_1 passing through the $1.5\text{ k}\Omega$ resistor and thus find the open circuit voltage V_{oc} .

KVL around the orange supermesh:

$$(2i_x) - [500\Omega]i_x - [2500\Omega]i_1 - [1500\Omega]i_1 = 0 \quad \Rightarrow \quad -4000i_1 - 498i_x = 0$$

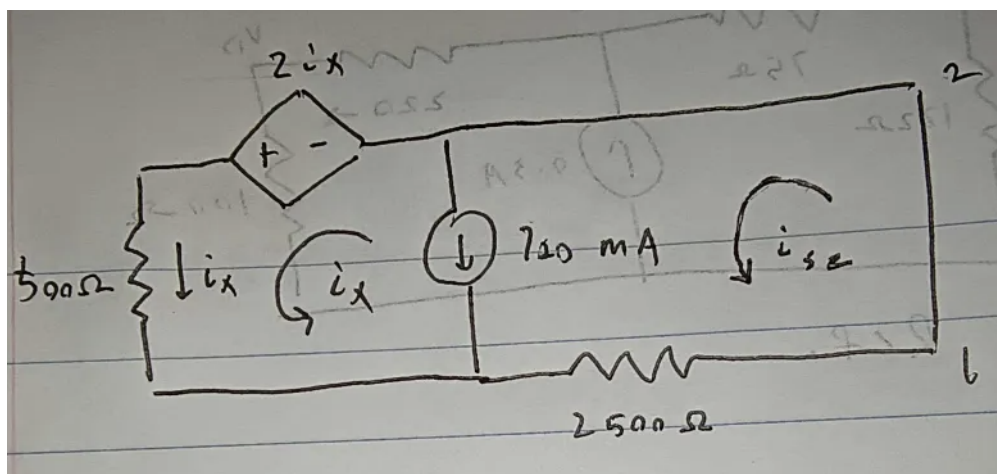
Supermesh equation:

$$i_1 - i_x = 700 \times 10^{-3} \text{ A}$$

Thus, $i_1 = 0.0775 \text{ A}$ and $i_x = -0.6224 \text{ A}$. Therefore, the open circuit voltage is:

$$V_{oc} = [1.5 \text{ k}\Omega]i_1 = \boxed{116.25 \text{ V}}$$

Next, we find the short-circuit current i_{sc} . When the terminals of the load resistor are shorted, we can again apply mesh analysis. The short creates a path with essentially zero resistance in parallel with the $1.5 \text{ k}\Omega$ resistor, so the current flows through the short instead. As a result, no current passes through the $1.5 \text{ k}\Omega$ resistor.



KVL around the supermesh:

$$(2i_x) - [500 \Omega]i_x - [2500 \Omega]i_{sc} = 0 \quad \Rightarrow \quad -498i_x - 2500i_{sc} = 0$$

Supermesh equation:

$$i_{sc} - i_x = 700 \times 10^{-3} \text{ A}$$

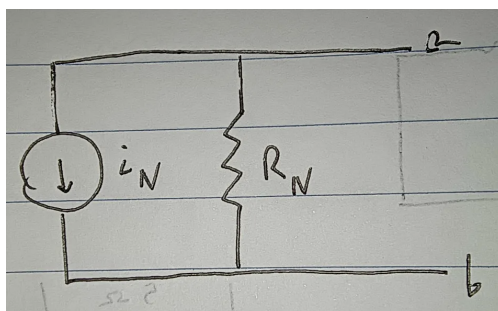
Thus, $i_{sc} = 0.1162 \text{ A}$ and $i_x = -0.5837 \text{ A}$.

Finally, we can find the Thevenin equivalent resistance:

$$R_{Th} = \frac{V_{oc}}{i_{sc}} = \frac{[116.25 \text{ V}]}{[0.1162 \text{ A}]} = \boxed{1000.43 \Omega}$$

The Norton equivalent is then:

$$\boxed{i_N = i_{sc} = 0.1162 \text{ A}, \quad R_N = R_{Th} = 1000.43 \Omega}$$



5.44

44. Referring to the circuit of Fig. 5.85, determine the Thévenin equivalent resistance of the circuit to the right of the dashed line. This circuit is a common-collector transistor amplifier, and you are calculating its input resistance.

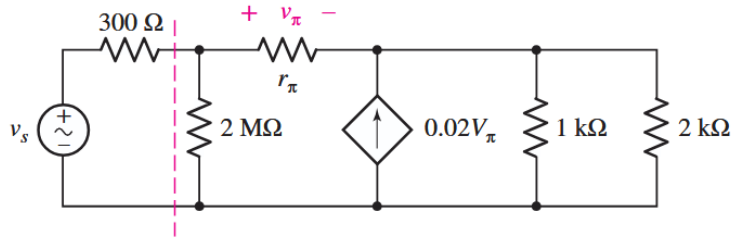


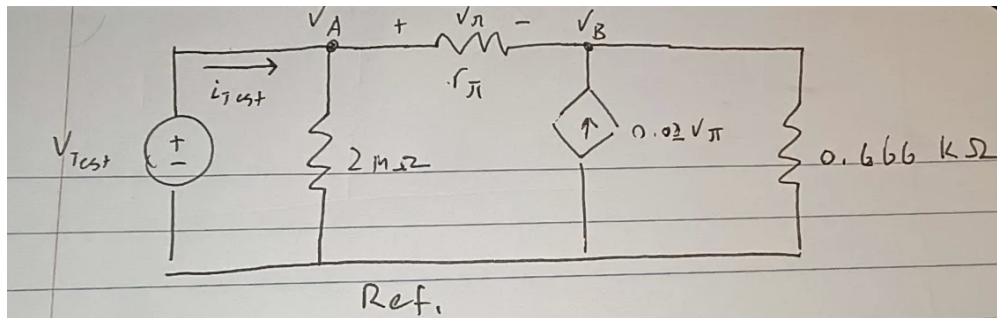
FIGURE 5.85

For this circuit, the dependent current source means that we cannot sum up the resistances to find the Thevenin equivalent resistance. We can either find the open circuit voltage V_{oc} and short circuit current i_{sc} or we can apply a test voltage source V_{Test} across the load terminals and find the resulting current i_{Test} to find the Thevenin equivalent resistance.

Note that because there are no *independent sources* in the circuit, both the open circuit voltage V_{oc} and short circuit current i_{sc} are zero. Thus, we cannot find the Thevenin equivalent resistance using the first method. So, we will use the test source method (turning off all independent sources to see the circuit's response; there are no independent sources in this case).

We can simplify the 1 k Ω and 2 k Ω resistors in parallel.

$$R = \frac{[1 \cdot 2 \text{ k}\Omega]}{[1 + 2 \text{ k}\Omega]} = 0.666 \text{ k}\Omega$$



$$V_{\pi} = V_A - V_B$$

$$V_{Test} = V_A$$

KCL at node A:

$$i_{Test} = \frac{V_A}{[2000 \text{ k}\Omega]} + \frac{V_A - V_B}{r_{\pi}}$$

KCL at node B:

$$\frac{V_A - V_B}{r_{\pi}} = [-0.02V_{\pi}] + \frac{V_B}{[0.666 \text{ k}\Omega]} \implies \frac{V_A - V_B}{r_{\pi}} = [-0.02(V_A - V_B)] + \frac{V_B}{[0.666 \text{ k}\Omega]}$$

$$V_A \left(0.02 + \frac{1}{r_\pi} \right) + V_B \left(-\frac{1}{r_\pi} - 0.02 - \frac{1}{0.666} \right) = 0$$

$$V_B = V_A \frac{\left(0.02 + \frac{1}{r_\pi} \right)}{\left(\frac{1}{r_\pi} + 0.02 + \frac{3}{2} \right)}$$

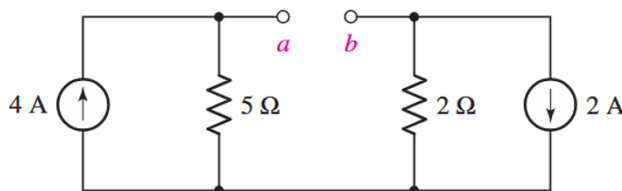
Plug this into the equation for i_{Test} to find the ratio $R_{Th} = \frac{V_{\text{Test}}}{i_{\text{Test}}} = \frac{V_A}{i_{\text{Test}}}$:

$$i_{\text{Test}} = V_A \left(\frac{1}{[2000 \text{ k}\Omega]} + \frac{1}{r_\pi} \right) - \frac{1}{r_\pi} \left[V_A \frac{\left(0.02 + \frac{1}{r_\pi} \right)}{\left(\frac{1}{r_\pi} + 0.02 + \frac{3}{2} \right)} \right]$$

$$\frac{1}{R_{Th}} = \frac{i_{\text{Test}}}{V_A} = \left(\frac{1}{2000} + \frac{1}{r_\pi} \right) - \frac{1}{r_\pi} \left[\frac{\left(0.02 + \frac{1}{r_\pi} \right)}{\left(\frac{1}{r_\pi} + 0.02 + \frac{3}{2} \right)} \right]$$

$$R_{Th} = \left[\left(\frac{1}{2000} + \frac{1}{r_\pi} \right) - \frac{1}{r_\pi} \left[\frac{\left(0.02 + \frac{1}{r_\pi} \right)}{\left(\frac{1}{r_\pi} + 0.02 + \frac{3}{2} \right)} \right] \right]^{-1}$$

- 5.51** 51. With reference to the circuit of Fig. 5.91, (a) find the Thévenin equivalent of the network defined by terminals *a* and *b*; (b) determine the Norton equivalent. (c) What resistor value connected between the open terminals results in maximum power transfer from the network?



■ **FIGURE 5.91**

Ref.

(a) Start by finding the open circuit voltage V_{oc} across nodes *a* and *b*, keeping track of polarity.

$$V_a = [5 \Omega][4 \text{ A}] = 20 \text{ V}$$

$$V_b = [2 \Omega][-2 \text{ A}] = -4 \text{ V}$$

$$V_{Th} = V_{oc} = V_a - V_b = 20 \text{ V} - (-4 \text{ V}) = \boxed{24 \text{ V}}$$

The Thevenin equivalent resistance after turning off the independent sources is:

$$\boxed{R_{Th} = 5 \Omega + 2 \Omega = 7 \Omega}$$

(b) Using the Thevenin equivalent, we just source transform to find the Norton equivalent:

$$\boxed{i_N = \frac{V_{Th}}{R_{Th}} = \frac{24 \text{ V}}{7 \Omega} = 3.428 \text{ A}, \quad R_N = R_{Th} = 7 \Omega}$$

(c) The maximum power transfer occurs when the load resistance R_L is equal to the Thevenin equivalent resistance R_{Th} :

$$R_L = R_{Th} = 7 \Omega$$

5.53 53. Select a value for R_L in Fig. 5.93 such that it will absorb maximum power from the circuit.

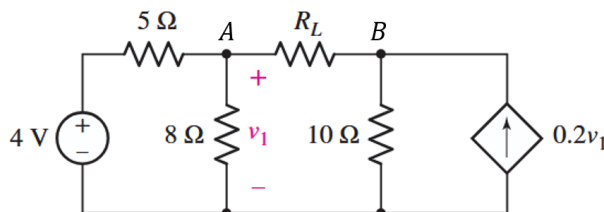


FIGURE 5.93

We want to find the Thevenin equivalent resistance. Since there is a dependent source, we either have to find the open circuit voltage V_{oc} and short circuit current i_{sc} or we can apply a test source. We will use the open circuit voltage and short circuit current method.

Find the voltage at node A by using voltage division:

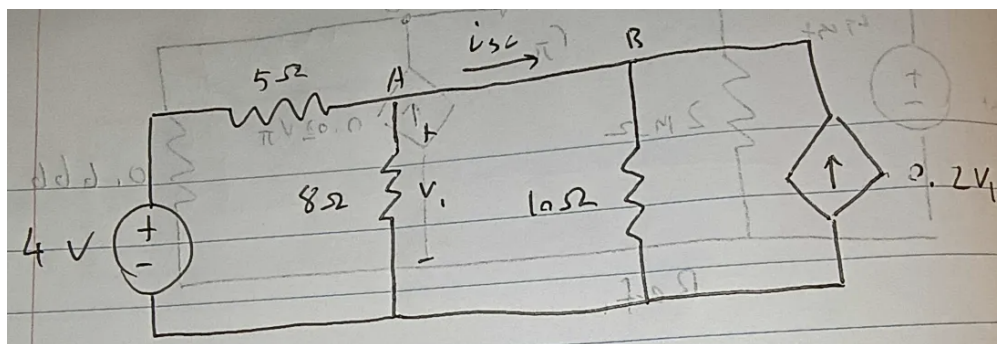
$$V_A = \frac{[8 \Omega]}{[5 + 8 \Omega]} [4 \text{ V}] = 2.4615 \text{ V}$$

Find the voltage at node B by using Ohms law ($V_1 = V_A$):

$$V_B = [10 \Omega](0.2V_1) = 0.2[10 \Omega][2.4615 \text{ V}] = 4.923 \text{ V}$$

$$V_{oc} = V_A - V_B = [2.4615 \text{ V}] - [4.923 \text{ V}] = -2.4615 \text{ V}$$

Now we add a short across nodes A and B to find the short circuit current i_{sc} :



KCL at node A & B supernode ($V_1 = V_A = V_B$):

$$\frac{4 - V_A}{[5 \Omega]} = \frac{V_A}{[8 \Omega]} + \frac{V_A}{[10 \Omega]} - (0.2V_1) \implies V_1 = 3.5555 \text{ V}$$

KCL at node B:

$$i_{sc} = \frac{V_1}{[10 \Omega]} - (0.2V_1) = \frac{3.5555 \text{ V}}{[10 \Omega]} - (0.2)(3.5555 \text{ V}) = 0.35555 \text{ A} - (0.7111 \text{ V}) = -0.35555 \text{ A}$$

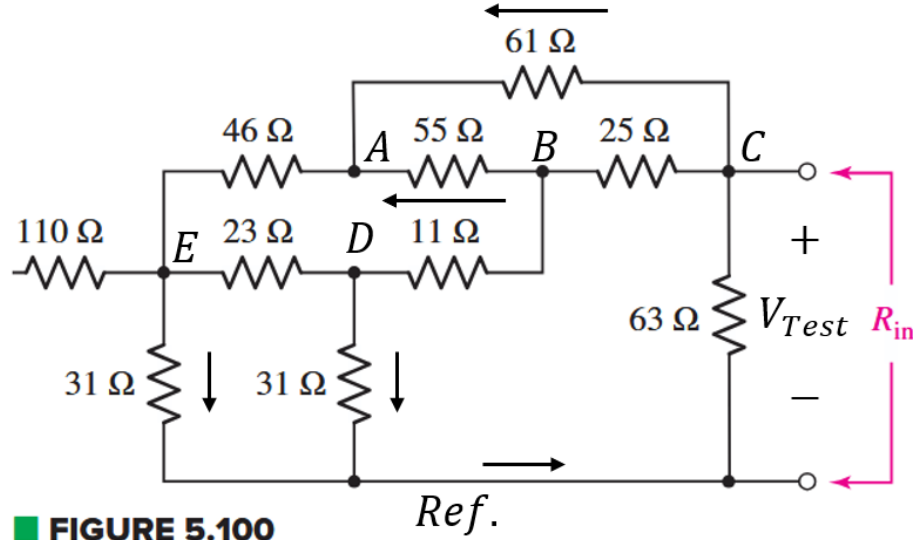
Thus, the Thevenin equivalent resistance is:

$$R_{Th} = \frac{V_{oc}}{i_{sc}} = \frac{[-2.4615 \text{ V}]}{[-0.35555 \text{ A}]} = \boxed{6.923 \Omega}$$

The load that absorbs maximum power is when $R_L = R_{Th} = 6.923 \Omega$.

5.61

61. Calculate R_{in} as indicated in Fig. 5.100.



■ FIGURE 5.100

Ref.

Using a test source $V_{Test} = 1 \text{ V}$, we can find the current i_{Test} and thus the equivalent resistance of the circuit.

Node A:

$$\frac{V_C - V_A}{[61 \Omega]} + \frac{V_B - V_A}{[55 \Omega]} = \frac{V_A - V_E}{[46 \Omega]} \implies -0.056314V_A + 0.01818V_B + 0.01639V_C + 0.0217V_E = 0$$

Node B:

$$\frac{V_C - V_B}{[25 \Omega]} = \frac{V_B - V_A}{[55 \Omega]} + \frac{V_B - V_D}{[11 \Omega]} \implies 0.01818V_A - 0.14909V_B + 0.04V_C + 0.09091V_D = 0$$

Node C:

$$V_C = 1 \text{ V}$$

Node D:

$$\frac{V_B - V_D}{[11 \Omega]} = \frac{V_D - V_E}{[23 \Omega]} + \frac{V_D}{[31 \Omega]} \implies 0.09091V_B - 0.16665V_D + 0.04348V_E = 0$$

Node E:

$$\frac{V_A - V_E}{[46 \Omega]} + \frac{V_D - V_E}{[23 \Omega]} = \frac{V_E}{[31 \Omega]} \implies 0.02174V_A + 0.04348V_D - 0.09748V_E = 0$$

Thus, solving the system of equations: $V_A = 0.6010 \text{ V}$, $V_B = 0.5865 \text{ V}$, $V_C = 1 \text{ V}$, $V_D = 0.4016 \text{ V}$, $V_E = 0.3132 \text{ V}$.

Thus using KCL at node C:

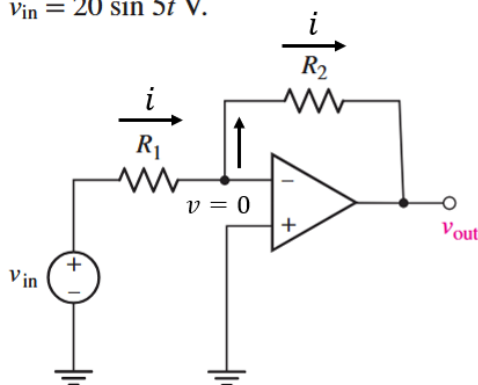
$$i_{\text{Test}} = \frac{V_C - V_A}{[61 \Omega]} + \frac{V_C - V_B}{[25 \Omega]} + \frac{V_C}{[63 \Omega]} = 0.03906 \text{ A}$$

Thus, the equivalent resistance of the circuit is:

$$R_{eq} = \frac{V_{\text{Test}}}{i_{\text{Test}}} = \frac{1 \text{ V}}{0.03906 \text{ A}} = \boxed{25.599 \Omega}$$

6.1

1. For the op amp circuit shown in Fig. 6.39, calculate v_{out} if (a) $R_1 = R_2 = 100 \Omega$ and $v_{\text{in}} = 5 \text{ V}$; (b) $R_2 = 200R_1$ and $v_{\text{in}} = 1 \text{ V}$; (c) $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 47 \text{ k}\Omega$, and $v_{\text{in}} = 20 \sin 5t \text{ V}$.



■ FIGURE 6.39

Op-amp rules: No current ever flows into either input terminal. (Current can flow at the output terminal!) There is no voltage difference between the two input terminals. It can either be a source or a sink of current.

(a) This is an inverting amplifier. We can find the relationship between the input voltage V_{in} and output voltage V_{out} by first using finding the current i by using the fact that the voltage at the inverting terminal is equal to the voltage at the non-inverting terminal (grounded). Thus, the voltage drop over R_1 must be equal to V_{in} :

$$V_{\text{in}} - R_1 i = 0 \implies i = \frac{V_{\text{in}}}{R_1}$$

Next, we can sum the voltage drops around the series connection between R_1 and R_2 to find the relationship between V_{in} and V_{out} :

$$V_{\text{in}} - R_1 i - R_2 i = V_{\text{out}} \implies V_{\text{out}} = V_{\text{in}} - (R_1 + R_2) i$$

Substituting the expression for i :

$$V_{\text{out}} = V_{\text{in}} - (R_1 + R_2) \frac{V_{\text{in}}}{R_1} = V_{\text{in}} \left(1 - \frac{R_1 + R_2}{R_1} \right) = -\frac{R_2}{R_1} V_{\text{in}}$$

$$V_{out} = -\frac{R_2}{R_1} V_{in}$$

Now we just plug in $R_1 = R_2 = 100 \Omega$, $V_{in} = 5 \text{ V}$

$$V_{out} = -\frac{100 \Omega}{100 \Omega} [5 \text{ V}] = \boxed{-5 \text{ V}}$$

(b) Plug in $R_2 = 200R_1$, $V_{in} = 1 \text{ V}$:

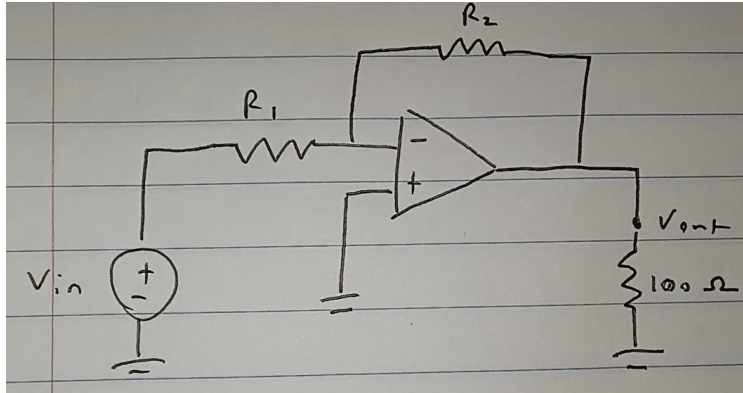
$$V_{out} = -\frac{200R_1}{R_1} [1 \text{ V}] = \boxed{-200 \text{ V}}$$

(c) Plug in $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 47 \text{ k}\Omega$, $V_{in} = 20 \sin(5t) \text{ V}$:

$$V_{out} = -\frac{47 \text{ k}\Omega}{4.7 \text{ k}\Omega} [20 \sin(5t) \text{ V}] = \boxed{-200 \sin(5t) \text{ V}}$$

6.2

2. Determine the power dissipated by a 100Ω resistor connected between ground and the output pin of the op amp of Fig. 6.39 if $v_{in} = 4 \text{ V}$ and (a) $R_1 = 2R_2$; (b) $R_1 = 1 \text{ k}\Omega$ and $R_2 = 22 \text{ k}\Omega$; (c) $R_1 = 100 \Omega$ and $R_2 = 101 \Omega$.



The voltage across the 100Ω is the same as V_{out} and the power dissipated is given as $P = V^2/R$.

(a)

$$V_{out} = -\frac{R_2}{2R_2} [4 \text{ V}] = -2 \text{ V} \implies P = \frac{(-2 \text{ V})^2}{100 \Omega} = \boxed{0.04 \text{ W}}$$

(b)

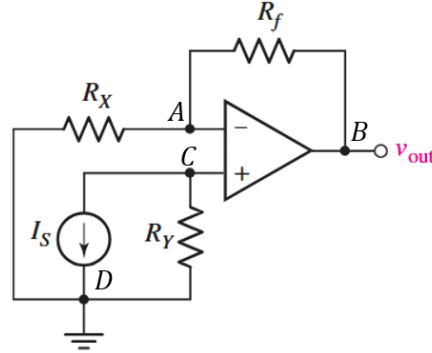
$$V_{out} = -\frac{22 \text{ k}\Omega}{1 \text{ k}\Omega} [4 \text{ V}] = -88 \text{ V} \implies P = \frac{(-88 \text{ V})^2}{100 \Omega} = \boxed{77.44 \text{ W}}$$

(c)

$$V_{out} = -\frac{101 \Omega}{100 \Omega} [4 \text{ V}] = -4.04 \text{ V} \implies P = \frac{(-4.04 \text{ V})^2}{100 \Omega} = \boxed{0.1632 \text{ W}}$$

6.16

16. For the circuit depicted in Fig. 6.46, calculate v_{out} if $I_s = 2 \text{ mA}$, $R_Y = 4.7 \text{ k}\Omega$, $R_X = 1 \text{ k}\Omega$, and $R_f = 500 \Omega$.



■ FIGURE 6.46

Using KCL at the non-inverting input to find the voltage there:

$$-I_s = \frac{V_C}{R_Y} \implies V_C = -I_s R_Y = -[2 \times 10^{-3} \text{ A}][4.7 \times 10^3 \Omega] = -9.4 \text{ V}$$

We can use the rule that the voltage at node A is the same as the voltage at node C to find the output voltage using KCL at node A (sum of current entering node A is zero, since it is an input terminal):

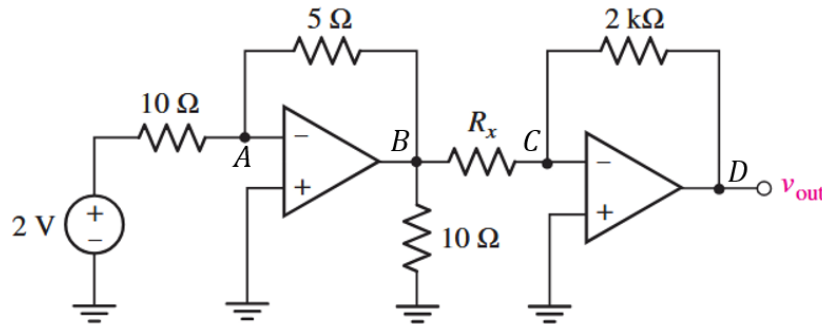
$$V_C = V_A$$

$$\frac{0 - V_A}{R_X} + \frac{V_A - V_{\text{out}}}{R_f} = 0 \implies V_{\text{out}} = V_A \left(1 + \frac{R_f}{R_X} \right)$$

$$V_{\text{out}} = [-9.4 \text{ V}] \left(1 + \frac{[500 \Omega]}{[1 \times 10^3 \Omega]} \right) = -14.1 \text{ V}$$

6.20

20. For the circuit of Fig. 6.48, determine the value of R_x that will result in a value of $v_{\text{out}} = 10 \text{ V}$.



■ FIGURE 6.48

Since the op-amp on the left is an inverting amplifier, we can find the output voltage at node B using the equation:

$$V_{\text{out}} = -\frac{R_2}{R_1} V_{\text{in}} \implies V_B = -\frac{[5 \Omega]}{[10 \Omega]} [2 \text{ V}] = -1 \text{ V}$$

We know that the voltage at node C is zero since the non-inverting input is grounded. Thus, we can use KCL at node C to find the output voltage in terms of R_x :

$$\frac{V_C - V_B}{R_x} = \frac{V_{\text{out}} - V_C}{[2 \text{ k}\Omega]} \implies R_x = -\frac{V_B}{V_{\text{out}}} \cdot [2 \text{ k}\Omega] = -\frac{[-1 \text{ V}]}{[10 \text{ V}]} [2 \text{ k}\Omega] = \boxed{0.2 \text{ k}\Omega}$$

6.25

25. For the circuit shown in Fig. 6.51, compute v_{out} if (a) $v_1 = 2v_2 = 0.5v_3 = 2.2 \text{ V}$ and $R_1 = R_2 = R_3 = 50 \text{ k}\Omega$; (b) $v_1 = 0$, $v_2 = -8 \text{ V}$, $v_3 = 9 \text{ V}$, and $R_1 = 0.5R_2 = 0.4R_3 = 100 \text{ k}\Omega$.

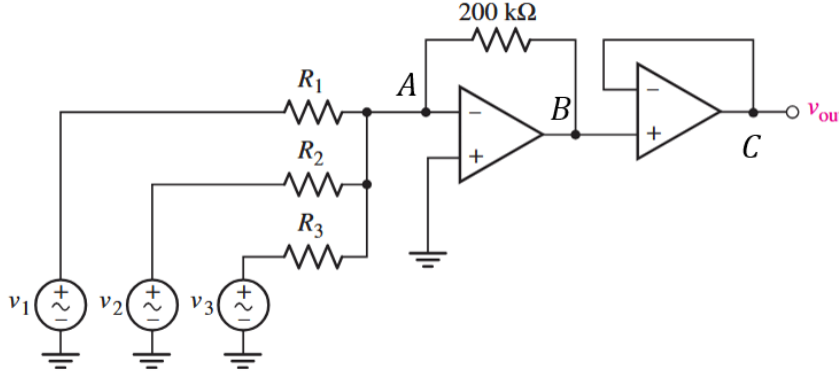


FIGURE 6.51

The op-amp on the left is an inverting summer circuit. The op-amp on the right is a buffer circuit. Thus, we can use the equations for a summer circuit to find the output voltage at node B:

$$V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right) \implies V_B = -\left(\frac{[200 \text{ k}\Omega]}{[R_1]}V_1 + \frac{[200 \text{ k}\Omega]}{R_2}V_2 + \frac{[200 \text{ k}\Omega]}{R_3}V_3\right)$$

The equation for a buffer circuit is

$$V_{\text{out}} = V_{\text{in}}$$

Thus, the output voltage is:

$$V_{\text{out}} = V_B = -\left(\frac{[200 \text{ k}\Omega]}{[R_1]}V_1 + \frac{[200 \text{ k}\Omega]}{R_2}V_2 + \frac{[200 \text{ k}\Omega]}{R_3}V_3\right)$$

(a) Plugging in $V_1 = 2V_2 = 0.5V_3 = 2.2 \text{ V} \implies V_2 = 1.1 \text{ V}$, $V_3 = 4.4 \text{ V}$ and $R_1 = R_2 = R_3 = 50 \text{ k}\Omega$:

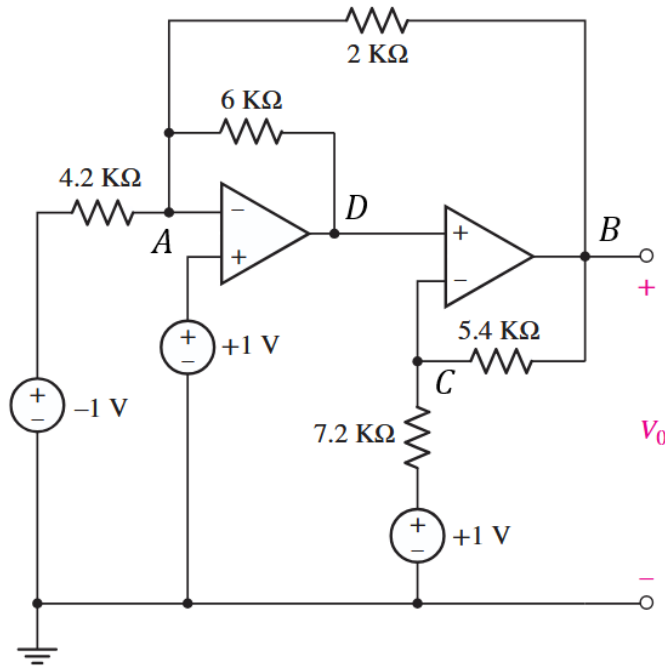
$$V_{\text{out}} = -\left(\frac{[200 \text{ k}\Omega]}{[50 \text{ k}\Omega]}[2.2 \text{ V}] + \frac{[200 \text{ k}\Omega]}{[50 \text{ k}\Omega]}[1.1 \text{ V}] + \frac{[200 \text{ k}\Omega]}{[50 \text{ k}\Omega]}[4.4 \text{ V}]\right) = \boxed{-30.8 \text{ V}}$$

(b) Plugging in $V_1 = 0$, $V_2 = -8 \text{ V}$, $V_3 = 9 \text{ V}$ and $R_1 = 0.5R_2 = 0.4R_3 = 100 \text{ k}\Omega \implies R_2 = 200 \text{ k}\Omega$, $R_3 = 250 \text{ k}\Omega$:

$$V_{\text{out}} = -\left(\frac{[200 \text{ k}\Omega]}{[100 \text{ k}\Omega]}[0] + \frac{[200 \text{ k}\Omega]}{[200 \text{ k}\Omega]}[-8 \text{ V}] + \frac{[200 \text{ k}\Omega]}{[250 \text{ k}\Omega]}[9 \text{ V}]\right) = \boxed{0.8 \text{ V}}$$

6.33

33. Calculate V_0 for the circuit in Fig. 6.55.



■ FIGURE 6.55

Sadly we will need to do nodal analysis to find the output voltage. Starting with the left op-amp, the voltage at node A is 1 V since the non-inverting input is 1 V.

Using KCL at node A ($V_A = 1$ V):

$$\frac{[-1 \text{ V}] - V_A}{[4.2 \text{ k}\Omega]} = \frac{V_A - V_D}{[6 \text{ k}\Omega]} + \frac{V_A - V_B}{[2 \text{ k}\Omega]} \implies 0.90476V_A - 0.5V_B - 0.16666V_D = -0.238095$$

To find V_D we can use KCL at node C ($V_C = V_D$):

$$\frac{V_B - V_C}{[5.4 \text{ k}\Omega]} = \frac{V_C - [1 \text{ V}]}{[7.2 \text{ k}\Omega]} \implies 0.185185V_B - 0.324074V_D = -0.138888$$

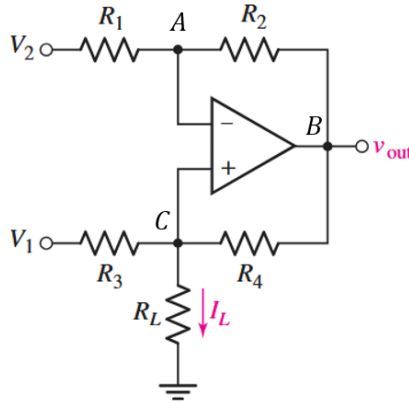
Thus, $V_B = 1.8000$ V and $V_D = 1.4571$ V.

The output voltage is the same as the voltage at node B, so

$$\boxed{V_{\text{out}} = V_B = 1.8 \text{ V}}$$

Note: The -1V source makes it a bit trippy but stick with convention that there is a voltage drop across the 4.2 kΩ resistor along the direction of current, thus the potential difference is $(-1 \text{ V}) - V_A$.

- 6.56** 56. For the circuit depicted in Fig. 6.64, known as a Howland current source, set $V_2 = 0$, $R_1 = R_3$, and $R_2 = R_4$; then solve for the current I_L when $R_1 = 2R_2 = 1 \text{ k}\Omega$ and $R_L = 100 \Omega$.



■ **FIGURE 6.64**

Given $V_2 = 0 \text{ V}$ and $R_1 = 1 \text{ k}\Omega$, $R_2 = 0.5 \text{ k}\Omega$, $R_3 = 1 \text{ k}\Omega$, $R_4 = 0.5 \text{ k}\Omega$, $R_L = 0.1 \text{ k}\Omega$., we can write the following nodal equations:

$$V_A = V_C$$

Node A:

$$\frac{V_2 - V_A}{R_1} = \frac{V_A - V_B}{R_2} \implies V_C \left(\frac{1}{R_2} + \frac{1}{R_1} \right) - \frac{V_B}{R_2} = 0 \implies V_B = 1.5V_C$$

Node C:

$$\frac{V_B - V_C}{R_4} = \frac{V_C - V_1}{R_3} + \frac{V_C}{R_L} \implies \frac{V_B}{R_4} - V_C \left(\frac{1}{R_4} + \frac{1}{R_3} + \frac{1}{R_L} \right) = -\frac{V_1}{R_3}$$

Substituting V_B into the equation for node C and plugging in the values for the resistors:

$$3V_C - 13V_C = -V_1 \implies V_C = \frac{V_1}{10}$$

Thus the current I_L is:

$$I_L = \frac{V_C}{R_L} = \frac{V_1/10}{[100 \Omega]} = \boxed{0.001V_1}$$